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The impact of AI on the economic development of the gaming industry

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Abstract

The aim of the article is to show how the implementation of artificial intelligence has influenced the development of the gaming industry, especially in terms of economy. The growing demand for games, the way of spending free time confirm the changing attitudes of society, and the growing needs, interests and expectations of customers. Among other things, these phenomena meant that in 2024 the gaming industry recorded an increase in the number of paying players by 5%, generating USD 1.50 billion in income, and the number of mobile games by 3%. This indicates the need to implement modern technologies in the industry to adapt to the trends of new stakeholders. The transformation is focused on broadly understood technological changes, in particular supported by the capabilities of artificial intelligence, which defines new directions of development of this industry and significantly affects efficiency and the labor market [1]. The trends presented in the article indicate emerging development opportunities both in the technical and technological aspect, as well as in the organizational or personal aspect of the gaming industry.

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1. Introduction

Computerization and globalization have significantly changed the needs of market stakeholders. Consumer behavior, especially of the young generation focused on "cybernetic life", has increased the demand for devices with a high level of technological advancement, which are used in everyday life, e.g. maintenance-free vacuum cleaners, cooking devices (Thermomix), mobile phones, etc. The change in the way everyday activities are performed has caused a change in the attitude to spending free time. The quality of life, and in particular the "work-life balance" created the opportunity to find a new way to relax, relieve stress or spend time pleasantly while playing computer games, VOD, streaming platforms, etc. [2-4]. The technology used in PC, phone and console games has allowed for social integration and meeting the expectations of players to a slow degree, and the high competitiveness of entities creating games and the small number of new games meant that the gaming market did not develop too quickly. Direct interaction of players did not guarantee new experiences, which is why online meetings are becoming increasingly popular, which was also contributed to by the COVID-19 pandemic [5].

New expectations and requirements of players towards game producers, and in particular in the aspect of increasing the attractiveness of the unpredictability of the game, have stimulated the demand for artificial intelligence. The implementation of artificial intelligence in the game creation process and its integration aims to diversify games, personalize the players' experience, improve manufacturing processes (increasing industrial production) while contributing to economic development and technical progress, increase demand and supply in the gaming industry [6]. In the gaming industry, technological progress provides great opportunities for change, both generating new opportunities and challenges, as well as threats, e.g., cybersecurity or compulsive behavior among adolescent online gamers [7]. In order to fully exploit the economic potential of the transformation of the gaming market, which is creating a "new" way of spending free time (entertainment), it is necessary to understand the impact of artificial intelligence (AI) on this industry [8,9].

The aim of the article is to attempt to demonstrate the impact of artificial intelligence on economic results in the gaming industry. To achieve the goal, the following hypothesis was put forward:

- H0: implementing AI in the game production process significantly contributed to improving the efficiency of the gaming industry,
- H1: despite the need to introduce organizational changes to the work process resulting from the implementation of AI, profitability indicators increased.

2. Technological transformation - selected components

The use of artificial intelligence is introducing significant changes in various sectors of the world's economies, including the gaming industry. The gaming market is one of the most active markets, which is why the expectations and requirements of players become a determinant of the development of games for entities creating them. One of the basic factors of change was the growing awareness of technical and technological innovations and the need to implement them. The result of this is the results in the sphere of games, as there appeared: [10,11]

- Virtual reality, where the player is transported into a "virtual world" full of possibilities, temptations, challenges, within which he exceeds the limits of his abilities (in the real world). Creating realistic and interactive experiences with sensory engagement includes high graphic fidelity, reality control and high quality sound (spatial).
- Augmented reality blurring the boundaries between reality and the virtual world through the use of cameras or AR glasses, guaranteeing the player "real" experiences in a fantasy world.
- Artificial intelligence allows for increased realism in games, increasing the potential of the game and the player's feelings in line with their character.
- A gaming space called platforms, providing opportunities for integration for players and the ability to engage in a "virtual world".

Also forced by changes resulting from technical and technological implementations were the reorganization of workplaces and a change in the approach to the employee and his needs [12,13]. This became particularly visible in the approach to performing work, currently based on project activities, which has been made more flexible and based on clearly defined performance indicators [14,15]. The already mentioned necessity to implement artificial

intelligence, among others, allowed for the automation of game creation processes (replacing many routine tasks currently performed by people) or analysis (testing), as well as marketing and distribution. This allows, for example, programmers and designers to focus on the visionary, creative, and strategic aspects of the games they create. For example, AI algorithms can create realistic game worlds, generate content, and even design games based on players' preferences [16].

Taking such actions has led to entrepreneurs increasingly using artificial intelligence for basic tasks, especially those performed by the majority of the human workforce in many areas. Research confirms that artificial intelligence can reduce human involvement to a minimum, and decision-making can be left solely to managers, directors or administrators. This is visible in machine learning, which involves creating models that learn from data and make decisions. Machine learning is divided into three main types: [17,18]

- Supervised Learning: Involves training models based on input and output data.
- Unsupervised Learning: Models learn on their own, looking for patterns in unlabeled data.
- Reinforcement Learning: The model learns by interacting with the environment and adapting its actions to maximize rewards.

With machine learning, computers can recognize patterns, classify data, and predict outcomes without the need for detailed programming. Therefore, it is believed that the implementation and use of AI in workplaces requires appropriate supervision on the one hand to maximize its benefits, but on the other hand to minimize the disadvantages and threats to the performance of work by employees [19,20].

However, it is worth remembering that the implementation and use of AI in game development processes must be supervised and properly secured, in particular in terms of [21,22]:

- cybersecurity controls needed to protect AI systems and ensure that they are prepared to respond and act in the event of disruptions;
- implementation, covering the preparatory, implementation and monitoring stages throughout the life cycle;
- Control of AI systems with interference/interaction with employees within the implemented processes, as well as at the interface between technology and business operations;
- Control of data sharing for AI and methods of their processing;
- Control of risk tolerance for AI;
- Control of compliance of AI implementation with adopted procedures, company policy, applicable law.

In summary, actions resulting from technological transformation, based on artificial intelligence, force employees to think critically and gain specialist knowledge, which, combined with employee skills and AI performance, generates increased added value for the organization [23,24].

3. Organizational and personal transformation - selected components

Changing the conditions and method of performing work as part of the process of improving the provision of work in the gaming industry is mainly based on creating an appropriate IT infrastructure (including work tools, e.g. computer hardware, software, workstations, etc.) with the required parameters regarding both throughput and design capabilities as well as employee expectations. An additional element of the change is the modification of the approach (perception) to the employee, in particular from the point of view of "soft" aspects. As indicated by research on technological innovations in the interpersonal area, significant impact is visible in the areas of the need to increase skills, competences and knowledge in the field of: [25-28]

- Solving problems in the areas of ICT and analytics.
- Acquiring and improving skills related to programming, design, creating game graphics.
- Creativity - creating innovative and engaging solutions in games, mechanics or graphics, or sounds (music).
- Teamwork, especially cooperation in interdisciplinary and international teams, on common goals.
- Effective and efficient communication.
- Using a programming language (Java, Swift, C#, JavaScript).
- Implementing and using artificial intelligence.
- Data analysis covering all input and output data, e.g. about players, in order to improve games and create more personalized experiences.

The change in the form of work and the approach to its performance resulting from technological transformation has also led to a change in the demand for employees and their qualifications. There has been an increased interest in employees, often independent, which results from a different division of labor, who are flexible in terms of working time, the way of performing work (mobility, remote work) or communication (reporting, interactions). This became visible during the emergence and duration of the COVID-19 pandemic, providing the opportunity to work remotely in interaction with other employees (creators) domestic and foreign [29,30]. It is worth noting that changes in the way of working, especially those related to the ability to learn quickly or adapt to new technologies, do not always bring the expected results, which has resulted in the loss of jobs or the need to increase the costs associated with improving qualifications [31].

The research results confirm this thesis, indicating that automation and AI are deepening social inequalities, leading to job losses in low-skilled sectors. Research from 2023 compared to 2022 showed that people working in the NLP and TensorFlow industries (programmers using a given language) are particularly at risk, as there has been a decrease in demand for employees:

- (NLP) junior by 48.3%, mild 55.3% and senior 53.4%;
- (TensorFlow) junior 50% and senior 62.2%.

A similar phenomenon occurs with game testers, 3D animators. The need to improve qualifications and skills through training or courses to improve qualifications is becoming justified. For example, programmers are focused on acquiring skills in Deep learning, a type of machine learning that uses multi-layered neural networks to model complex patterns in big data [32]. It should also be noted that a sudden change in the nature of work, increased employer expectations, and growing customer requirements, especially related to the implementation and use of artificial intelligence at work, may negatively impact an employee's mental health. According to the APA Institute (2023), as many as 51% of respondents indicated that working with artificial intelligence has a negative impact on mental health, as some (all) professional duties will become obsolete. Only 29% of respondents had a different opinion [33]. Constant stress resulting from the need to use AI and fear of losing their job make employees ineffective. Remote work and flexible forms of employment can also lead to social isolation and mental health problems. Employees may additionally feel a negative stigma, pressure related to constantly adapting to new technologies and developing new skills. It is important for employers and governments to create an environment that supports the mental health and well-being of employees, which should include:

- Offering mental health support programs: Access to counseling and other mental health services can help employees cope with stress and anxiety.
- Creating a welcoming atmosphere for employees: Promoting a culture of openness and respect can help employees feel comfortable and supported.
- Promoting work-life balance: Encouraging employees to take breaks and vacations and to maintain a healthy work-life balance can help them avoid burnout.

Despite existing threats and limitations, changing the organization of work creates many opportunities for both employees and the company, e.g. in creating new jobs requiring programming skills, machine learning and the use of artificial intelligence, which is confirmed by new jobs for specialists in the area of e.g. 3D programming (3D animators), testers, directors (game scriptwriters), etc. The demand for such professions, and in particular specialists in these areas, is high and constantly growing.

4. Management transformation - selected components

Implementing changes in an organization always brings with it a change in the perception of the future of the organization. Implementing new technical and technological solutions, and in this case artificial intelligence, to the business processes of the organization will always result in changes in the supporting and management processes. A new tool correlating with various processes causes a change in the approach to management. One of the effects (solutions) of such evolution, resulting from organizational changes, is the creation (increase) of the possibility of implementing selected tasks faster, easier, more effectively through the use of outsourcing. Outsource the production of individual game elements to countries with lower costs, maximizing profits. This may result in, on the one hand, a loss of jobs in developed countries, and on the other, development in countries with lower production costs [34]. In less developed countries, there may be a sudden increase in building intellectual capital in employees, in line with the

requirements and expectations of gaming stakeholders. Trends indicate growing interest in the Asian countries in the gaming industry, both in testing and generating ideas. The number of players is also growing. In 2024, the number of players increased (compared to 2023) by 4% in the Asia-Pacific region (reaching 1.809M), by 8.2% in the Middle East & Africa (559M), by 2.9% North America (244M), by 5.6% Latin America (355M) and by 2.4% in Europe (335M) [35]. This confirms the thesis that globalization creates opportunities to enter markets and enables international cooperation with various entities, in this case creating and testing projects, the effect of which are innovative games that meet the expectations of players. The development trend is also visible in the area of games, i.e. E-sports, competitive games, social gaming and indie gaming. For example, the increase in players in Esports was 9.6% (annualized 2020 to 2019) giving a value of 435.9M players; 8.7% (in 2022 to 2021) constituting 474.0M players. In 2024, the number of players increased to 577.2M [36]. This shows how dynamically this area of games, the gaming industry, is developing, the effect of which is a transformation encompassing the effect of generational (consumer) changes, globalization, computerization, technological progress, including artificial intelligence and even the COVID-19 pandemic.

Research has shown that the number of players in the world reached 3.42 billion in 2024, giving an increase of 4.5% by 2023. This situation is due to the growth of PC players, who had the strongest increase (by 2023) of as much as 3.9%. Other players had a lower increase, i.e. console players 2.3% and mobile players 3.5% [36]. This means that if demand is growing (gamers are most likely to play in their free time, as indicated by the significant increase in mobile gamers), then supply must offer products that meet existing needs and generate new ones (game producers must adapt their products to the technological capabilities of players (and vice versa) and create an increasingly large offering package). That is why it is so important that the changing nature of work in the gaming industry, especially with the use of AI, is adapted to the expectations of the employee, the organization, in order to solve all adversities and limitations and ensure lasting and sustainable development [37,38]. It is necessary and essential that with such a rapidly changing environment and growing competitiveness and customer expectations, entities creating games create and provide services at a satisfactory, repeatable level of quality [39,40]. It is important to state that quality is one of the main parameters of the competitiveness and development of an organization. Enterprises focused on improving quality in all areas of the organization, including the quality of work [41-44], will contribute to improving working conditions, creating potential for innovation, new products or services, increasing job satisfaction, indirectly influencing the shaping of the quality of life. Research has shown that improving the quality of work is most often associated with implemented innovations and the effects of these activities ensuring employee well-being and protection of the working environment [45,46]. The employer's new perspective on the work environment may result in a greater demand for specialist employees, whose supply on the labor market will be decreasing. This labor market phenomenon, plus the possibility of carrying out work in various forms, means that entrepreneurs are increasingly beginning to appreciate (enhance) employees financially. According to data from the No Fluff Jobs report (2022), Java programmers could count on gross earnings in the case of a B2B contract at the level of PLN 16-21.8 thousand, and employment contracts in the range of PLN 12-17.5 thousand. Then, people working in Swift technology earned in the range of PLN 15-21 thousand for B2B and PLN 13-17 thousand for an employment contract. The next group of the best-paid employees were employees using C# technology, who received an average of PLN 14.7-20.2 thousand for B2B and PLN 12.5-19.3 thousand for an employment contract. Android specialists could count on salaries in the range of PLN 14.3-20 thousand for B2B and PLN 12.2-17 thousand for an employment contract, and C++ specialists on slightly lower ones – between PLN 14 and 20 thousand for B2B and PLN 12-17 thousand for an employment contract. In the case of JavaScript, salaries ranged from PLN 13-18.9 thousand for B2B and PLN 12-17 thousand for an employment contract [47].

This changing force of the impact of dependencies creates a space of opportunities for specialists from various fields of the gaming industry, such as programmers, testers, designers, graphic designers, musicians, Big Data specialists, etc. In other words, a very active area is being created, activating various sources of financing, the labor market, employee potential, the activities of entrepreneurs, who must adapt the infrastructure and working conditions for employees (resulting from changing expectations) and also create jobs for e.g. analysts, because the importance of data and database analytics is increasing. Employers should also focus on developing employee potential using dedicated domestic and foreign online courses, specialist literature on the subject, active participation in workshops, seminars, competitions or conferences [48]. The exchange of knowledge and experience can build new relationships between market stakeholders, as well as identify and assess their capabilities in the context of new trends in the gaming

industry. Enterprises need to generate and manage data used to assess and understand player behavior, evaluate trends, in order to optimize player solutions and decisions [38,49]. It is important that entrepreneurs focused on using AI in production, management or support processes are guided by the principle of artificial intelligence risk management. Within this area, they should: [50]

- Define a risk mitigation strategy in line with legal acts – taking into account business goals in the field of AI, defining threats and possibilities of AI-related risks)
- Define and conduct risk supervision – indicate the responsibilities and scope of supervision
- Create and implement procedures, regulations and other acts to manage or eliminate the occurrence of risks related to artificial intelligence, determine the possible impact of AI impact, determine the path of escalation of undesirable phenomena
- Respond to incidents – create a method for reporting and identifying phenomena and determine who can solve them, supervise them at the job position level
- Monitor and study the impact – measure the impact, frequency, etc.
- Reporting - reporting the occurrence of the phenomenon to management. It is important for the success of the organization to set priorities, parameterize phenomena, improve processes will allow for the appropriate use of AI algorithms in various areas of the organization.

Additionally, the employer must effectively improve employees, providing them with the opportunity for periodic training, because both the organization and AI, as part of the work performed, develop. Entrepreneurs should therefore be open to changes and ready to adapt to new challenges, both employee and technological [51,52]. As the research results indicate, employers should pay attention to the psychological aspect, which is strongly visible in the gaming industry, when creating safe working conditions [7]. The ability to manage based on job security and stability allows employees to take care of their mental health in order to cope with stress and anxiety related to the changing nature of work and increasing expectations. Despite striving to increase employee work efficiency, entrepreneurs should strive to create a balance between work and private life [53,54].

To sum up, it should be pointed out that all actions taken by entities creating games aimed at, among others: creating new knowledge, jobs, developing artificial intelligence, deeper and comprehensive data analysis, increasing cybersecurity or creating and promoting digital marketing, must always generate costs, and their products must generate specific financial benefits. In 2024, research conducted by IDC indicated that for every \$1 invested in new AI technology in the gaming industry, a return of \$3.50 was received [55]. This means that investing in AI in the gaming industry pays off. At the same time, if we assume that the forecast for the global growth of the live gaming market in 2025-2030 will be as follows (in US\$ billions): [56]

- 2025 will reach a value of 12.3,
- 2026 will reach a value of 13.2,
- 2027 will reach a value of 14.2,
- 2028 will reach a value of 15.4,
- 2029 will reach a value of 17.3,
- and 2030 will reach a value of as much as 18.7.

We can therefore confirm (at the previously indicated level of return) that the profitability of the impact of AI, as part of the technological transformation, on the development of the gaming industry is significant. The research results published in the Unity Gaming Report (2025) [50] also indicate that over 79% of the surveyed game developers positively assess the use of AI. They indicate this tool as a potential for developing industries and their creators.

5. Conclusion

In the last decade, socio-economic, geopolitical or atmospheric changes have not had such a large impact on the limitations related to globalization or technological development in the gaming industry. Also, the emergence of the COVID-19 pandemic and radical changes related to the way society (especially young people) spends its free time did not stop, but rather accelerated the changes related to the implementation of modern solutions for working on games. Both automation and the use of artificial intelligence (AI) in game production processes have revolutionized the gaming industry, influencing almost every aspect of game creation and distribution. The integration of new

artificial intelligence technologies has on the one hand, positively changed the player's world by increasing experiences and feelings, personalizing games, etc., and on the other hand, forced companies (game producers) to introduce significant changes in the way they provide, monitor work or motivate employees. It would seem that the introduction of changes related to the adaptation of the place, tools for work (organizational resources) may have a significant negative impact on the gaming industry market, but nothing is wrong. The appropriate use of AI in manufacturing processes and the creation of modern experiential solutions for players resulted in a faster and greater return on investment, and the scale of market development only complemented the scale effect of profitability, which is confirmed by the data contained in the study. The conducted considerations related to the impact of AI on the development of the gaming industry, supported by data from various areas shaping development opportunities, confirm the hypotheses and clearly indicate that the future of this industry will be the improvement and comprehensive possibilities of using artificial intelligence by game authors and producers. Especially in the area of implementing independent characters, a realistic world, the ability to make decisions and show feelings, creating significant interactions between man and machine.

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